

Introduction to Machine Learning

1. What is Machine Learning?

Machine learning is a branch of artificial intelligence that allows computers to learn patterns from data instead of being explicitly programmed. A model is trained on examples and then used to make predictions on new, unseen inputs.

Definition: A training set is the collection of labelled examples that a learning algorithm uses to fit the parameters of a model.

The quality of a model depends heavily on the quality and quantity of its training data. Noisy labels, missing values and biased samples all reduce accuracy.

2. Types of Machine Learning

2.1 Supervised Learning

In supervised learning every training example has a known target label. Common tasks are classification, where the target is a category, and regression, where the target is a number.

Example: Predicting whether an email is spam from its words is a supervised classification problem.

2.2 Unsupervised Learning

Unsupervised learning finds structure in unlabelled data. Clustering groups similar points together and dimensionality reduction compresses features while keeping most of the information.

2.3 Reinforcement Learning

In reinforcement learning an agent interacts with an environment and learns a policy that maximises cumulative reward through trial and error.

3. Training a Model

Step 1: Collect and clean the data. Step 2: Split it into training and test sets. Step 3: Fit the model on the training set. Step 4: Evaluate it on the test set.

$$\text{MSE} = (1/n) * \sum (y_i - \hat{y}_i)^2$$

Important: Never evaluate a model on the same data that was used to train it.

4. Review Questions

Q1. What is the difference between classification and regression?

Answer: Classification predicts a discrete category, while regression predicts a continuous numerical value.

Q2. Why do we need a separate test set?

5. Conclusion

In conclusion, machine learning turns data into predictive models, and careful evaluation is what makes those models trustworthy.

References

- [1] Mitchell, T. (1997). Machine Learning. McGraw-Hill.
- [2] Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.